

Solve Word Problems Involving Inequalities

Before solving inequalities, it is important to recognize some key words.

English Phrases Describing Inequality	Algebraic Expression
Greater than “...is more than...” “...is larger than...”	
Less than “...is fewer than...” “...is smaller than...”	
Greater than or equal to “...is not less than...” “...is at least...” “...cannot be less than...” “...a minimum..”	
Less than or equal to “...is not greater than...” “...is at most...” “...is not more than...” “...does not exceed...” “...a maximum...”	

Translate each English expression into an inequality. Use x as the variable.

- a) The speed limit on the Outer Ring Road is a maximum of 100 km/h.
- b) The minimum wage in NL is \$15.60 per hour.
- c) A person's systolic blood pressure is classified as high if it is above 130.
- d) A grade below 70% is considered a failing grade in the Veterinary Technician Program.

Problems containing words or phrases such as at least, at most, between, no more than and no less than often require an inequality rather than an equation.

Examples

Solve using inequalities.

You need to buy lamps and curtains for your apartment. You have a maximum of \$340 to spend. A set of curtains costs \$48, and each lamp costs \$35. If you buy 4 sets of curtains, what is the maximum number of lamps you can afford? Assume the prices include taxes.

A taxi charges a flat rate of \$2.75 plus \$0.50 per kilometre travelled. Determine how far a customer can travel if they want to spend less than \$10.

The cost, C , of producing x coffee cups is given by $C = 105x + 900$. The revenue, R , generated from selling x coffee cups is given by $R = 115x$. Determine the values of x for which the company makes a profit.

The length of a rectangle is 3 metres more than twice its width. If the perimeter of the rectangle is at least 30 metres, write and solve an inequality to determine the possible values of the width.

A tank initially contains 400 L of water. Water drains from the tank at a rate of 1.25 L/min.

- a) Write an expression for the amount of water remaining in the tank after t minutes.

- b) For how long will the tank contain more than 100 L of water? Set up and solve an inequality.